

Multi-Agent Systems: Final Project

DAVID JIMENEZ BARRERO. R0607535

KU Leuven. Master Artificial Intelligence
david.jimenezbarrero@student.kuleuven.be

1. INTRODUCTION

For this project I created a Multi-Agent system for solving the logistics of AGV's transporting goods on a warehouse. I propose and test a solution for coordinating the AGV's using 2 types of agents in a semi-decentralized fashion in which AGV's have an assign agent, while there is a central agent which coordinates a task allocation following a Dynamic Contract-net protocol.

Additionally, we propose a scheme for collision avoidance by sending messages from AGV to AGV.

1.1. Objectives

- Design a *task allocating* system for coordinating parcel pickup and delivery footed on an parametrized *energy function* which takes into account the distance to the parcel, time to expiration of the parcel, and other parcels.
- The chosen energy function should optimizes, given the proper tuning, the time for delivering a predefined number of parcels.
- Investigate the efficiency of the chosen system.
- Design and implement a Planning Prior to Coordination system (PPC) with agent-to-agent coordination for AGV collision avoidance.

1.2. Research Questions

The research questions based on the objectives are:

- Can a task allocation based a function of distance, expiry time, and other parcels, improve the results obtained by simple distance assignment?
- What is the effectiveness of an agent-to-agent coordination for collision avoidance?

Finally the **hypotheses** for the project are:

- By including in the allocation process, the distance and time information of all the parcels in range, the system is able to do a better allocation selection which corresponds to a decrease in the time for completion.

- The implementation of a PPC approach is effective and produces a small amount of errors (un-avoided collisions and deadlocks)

2. THEORY

2.1. Types of Agents

For this problem we propose using 2 different agents working together, namely: Vehicle agents (control the AGVs), and a central allocating agent (CAA). The vehicle agents' tasks is to pick up and deliver parcels from its current location to the destination. On the other hand, the CAA monitors the parcels allocation. These agents are independent, however they must coordinate between them to accomplish their goals. In this system, Vehicle agents communicate with the CAA to obtain the assignment of a parcel, and Vehicle agents communicate with other vehicle agents to coordinate collision avoidance.

2.1.1 System-Architecture

This constitutes a semi-decentralized structure. I refer to it a semi-decentralized as, although it it has a centralized architecture, the central agent provides a "guidance" to the other agents. This contrasts with a centralized planner that produces detailed plans for the underlying agents, as this removes the ability of the agents to generate their own plans, or generate any individual action for that matter - contrasting with the definition of an agent. Nevertheless, in this scenario (centralized planner), the central agent has a broad view of the agents and the environment, and thus is able to produce very efficient solutions [2]. Therefore, the approach proposed is not a heterarchy (explained briefly in [3]) nor fully centralized, as the CAA agent only guides the Vehicle agents by telling them which is the most convenient parcel, however how they plan to reach the parcel and take it to the goal is an autonomous behavior. This architecture can be seen in Figure 1, where the arrows represent communication.

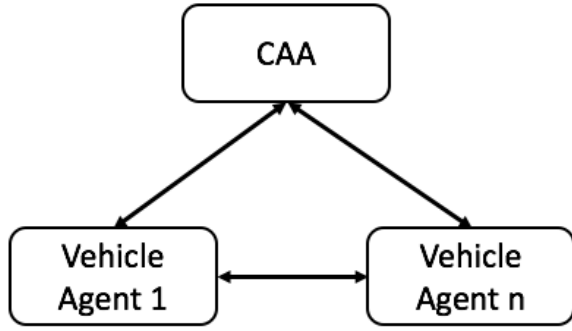


Figure 1: *Hierarchy of the multi-agent system.*

2.1.2 Agent Architecture

As mentioned before, I have 2 agents: Vehicle Agents and CAA. The Vehicle agents can be thought as a **Practical Reasoning** agent, where it perceives the environment. It then reasons how it *desires* to act, i.e., which parcels it would like to transport based on an energy function. It then settles on a particular intention, based on the information from the CAA. Based on this intention it plans how to execute this intention, particularly, based on the “Road Model” and the constraints it carries (based on the particular map). It then executes the intention.

Now here, some unplanned behavior can occur, as is to find other Vehicle agents in the way. In this scenario, the agent behaves as a **Reactive agent**, where the designer proposed a set of rules to deal with this situation. These are simple rules that do not require complex reasoning. They are intended to avoid a collision with other AGVs in the eventuality that the two are in path for collision.

This combination of the two architectures is then a **Hybrid Architecture**; in this architecture, the two behaviors take the form of *layers* of behavior. This agent is structured into a *horizontal layered* architecture. Here, the Reactive layer, as well as the Reasoning layer receive direct perceptual input and both can output actions. I.e., the Practical Reasoning layer receives information from the world (road network, CAA, and other agents) and executes actions based on the processing. The Reactive layer also takes in information about other AGV’s planned routes and can take action or not, based on its rules. This architecture can be seen in Figure 2.

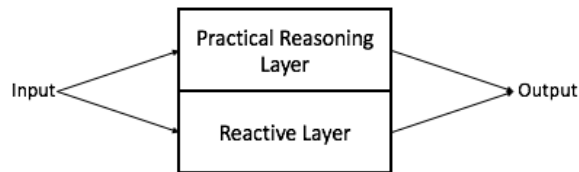


Figure 2: *Vehicle Agent: Hybrid architecture.*

Notice, however, that I do not include in this diagram I do not include a **control subsystem**, to coordinate which of the layers is in charge. This is because in the original design, this is not contemplated nor implemented; more comments on this in the Analysis section.

For the CAA, we have a Practical Reasoning architecture. This agent takes the knowledge about the AGV’s desires to transport parcels and what it *desires* to do: correspond a parcel with the best agent (dictated by the energy function). It then settles on some particular solution which becomes its intentions. It then executes them (informs the agents corresponding).

2.2. Dynamic Contract-Net Protocol

The Dynamic Contract-Net protocol is a coordination protocol for task allocation. It is based on the Contract-Net protocol. It proposes that a central agent “manager” recognizes a problem, offers the problem to agents which can potentially solve it, the agents “bid” in order to obtain the task, and finally the manager selects one winner agent to take on the task. The Dynamic Contract-Net protocol builds on top of this protocol as it allows re-allocation of tasks after assignation. I.e., once a task is assigned to an agent, in the eventuality that another agent emerges which is more suited for the task, or has a higher “bid”, the new agent is assigned for the task and the former agent becomes available again. Additionally, Dynamic Contract-Net protocol allows agents to reject proposed allocations, which is not possible in the regular Contract-Net.

In this project, we implemented a Dynamic Contract-Net protocol for allocation of parcels. Vehicle agents analyze the available parcels and make a “bid” to the CAA, based on which parcels they prefer. This bid is guided by the energy function proposed, which if smaller is more desirable. This bid is then based on the position of the agent with respect of the available parcels, their time of expire, and their delivery locations. The CAA agent analyzes all the incoming bids for a particular parcel and assigns the AGV with the smallest energy function for the particular parcel. Now, every time an AGV becomes available, it bids for a new parcel, if it has a smaller energy for an already assigned parcel, the CAA cancels the booking and reassigns the parcel.

2.3. Local Planning Prior to Coordination

The way the multi-agent system executes is a Local Planning Prior to Coordination. Here, each agent creates a plan locally, to achieve its goals or perform a task. Afterwards, the agents solve conflicts that may arise between to incompatible plans by agent-to-agent interaction or through a coordinating agent.

In this project, after an agent is assigned a parcel, it plans the shortest path to arrive to the pickup location, and then the shortest path to deliver it. **Sometimes, however, two agents must cross the same location**

at the same point, creating a spatial-resource conflict. To avoid this, agents “message” surrounding agents about their intentions, to which other agents might detect conflicts and act in order to solve them. In this sense, agents interact agent-to-agent to solve problems in a Local Planning Prior to Coordination.

2.4. Messaging

The agents’ communication protocol used in this project is based on the FIPA Agent Communication Language. The FIPA ACL is a standardization of Agent Communication Language. It is characterized for defining the types of messages and how they are interpreted by an agent’s context. FIPA has defined **performatives** that describe the purpose of the message, as well as the context it should be used.

In this project I used the performatives described in Table 1.

Performative	Context	Use
Inform	Pass Info.	Collision Avoidance
Request	Request Info.	Update Energy
Inform	Pass Info.	Confirm Booking
Inform	Pass Info.	Update Energy
Cancel	Negotiation	Cancel Booking
propose	Negotiation	Propose Booking

Table 1: FIPA performatives used; description and usage.

3. MULTI-AGENT SYSTEM DESIGN

In this section we concretely describe the design of the system; especially what has not been mentioned before in Section 2.

3.1. Design

3.1.1 Macro

The design of this system is composed in the macroscopic level by 2 types of agents, the CAA and the Vehicle agents. The goal of this design is to guide the Vehicle agents via a central agent CAA to achieve better solutions than a system of single heterarchical agents could produce. This additionally allows the system to have a **separation of concerns**, as the CAA focuses on managing the parcels, while the Vehicle agents focus on transporting the assigned parcels.

3.1.2 Micro

Individually, the Vehicle agents were designed with a Hybrid architecture so that they could reason for achieving their tasks, but also have the capacity to react to changes in the environment.

3.1.3 Energy Function

The energy function is the center of the project. It is designed to take into account the distance of an agent to the parcels, the time when the parcels expire, and additionally the **position the agent will be in after delivering the parcel**.

Therefore, in this energy function, it is **important to evaluate the distance to other parcels after delivery**. This way some parcels might be more convenient to pickup than others, solely on the fact that the delivery point is closer to other unassigned parcels. This is illustrated in Figure 3, where an AGV can pick up either Parcel A or Parcel B, however the delivery location of Parcel A is near Parcel C, while Parcel B’s delivery location is far. Then, although Parcel B is closer to the AGV, it might be more “profitable” to pickup Parcel A, as it leaves the AGV in position of being more productive later on.

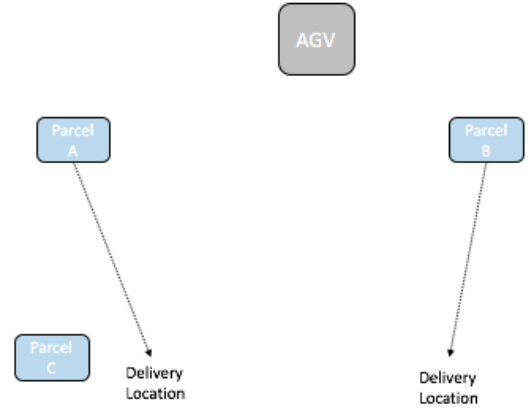


Figure 3: One AGV and 3 unassigned parcels.

The energy function $\mathcal{E}_i^n$ for a parcel i relative to the AGV n is proposed as:

$$\mathcal{E}_i^n = \alpha_1 d_i + \beta_1 (T_i^{exp}) + \alpha_2 d_j + \beta_2 (T_j^{exp}), \quad (1)$$

where d_i is the distance to the parcel i , T_i^{exp} is the time parcel i must be delivered, d_j is the distance to the closest parcel to the delivery point of parcel i (in this case, parcel j), T_j^{exp} is the time parcel j must be delivered; additionally the constants α , and β are proportionality constants. The lower the energy is, the more “profitable” the parcel is assumed to be.

Notice the importance of each of the quantities: the distances and the times of expiration, are dictated by the proportionality functions. Different combinations of these constants can severely affect the behavior of the agents.

The agent then sorts the parcels within reach by lowest to highest energy function, and sends this list to the CAA.

3.1.4 Negotiation

In this project, the negotiation occurs as each AGV sends the CAA the sorted list of the parcels. The evaluation by the CAA goes as follows:

1. The CAA evaluates the first proposal, if it is available, **the agent receives the booking**.
2. The CAA evaluates the first proposal, if it is unavailable, however the energy proposed is lower than the currently assigned, then **the agent receives the booking**. The other agent gets a cancellation.
3. The CAA evaluates the first proposal, if it is unavailable, and the energy proposed is higher or equal than the currently assigned, then **the CAA moves to evaluate the second best parcel for the agent in the same fashion**. The cycle restarts on the first item of this list.

In this way, the CAA guarantees that the parcels get assigned the lowest AGV possible. And the AGVs, if they cannot get the best energy parcel assigned, they might get the second or third (or more, depending on the number of AGVs and parcels present), which from the point of view of the AGV, it is still a profitable deal. In practice, I restricted the amount of times an AGV can change parcel to 2, as sometimes this dynamic canceling resulted in pairs of AGVs constantly changing between them the booking of a parcel.

3.1.5 Collision Avoidance

As mentioned before in Section 2, the Vehicle agents are composed of a layered architecture of Practical reasoning and Reactive behavior. The reactive behavior was designed explicitly for dealing with cases when AGVs might collide with this one, given they continue their planned trajectory. **The design of this system is based on a grid road map where all roads are traversable in both directions.**

It works by making the agent constantly send a message with its intended plan (n number of steps ahead) to other AGVs. When a near-by AGV picks up the message, it extracts the plan and analyzes it. The analysis is done by projecting the local AGV plan n steps into the future and checking for possible inconsistencies in both plans (same location $\pm m$ units of distance, at the same time). If there is an inconsistency, the reactive agent quickly takes over and sets route to the previous cross road. From there, it analyzes which alternate direction is better to take based on the location of its destination (pickup location or delivery). This is seen in Figure 4.

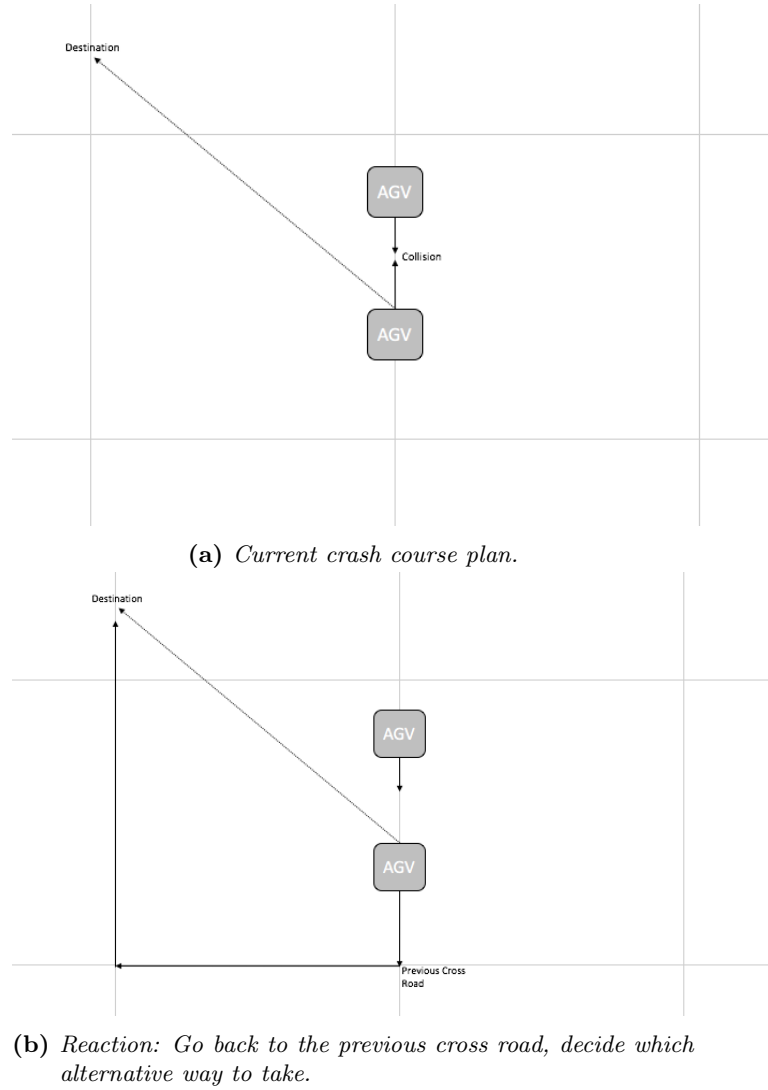


Figure 4: Process of a collision avoidance.

For this to work, it is required that $m \leq n \leq \text{size of each tile}$ (in terms of steps required to traverse it). This entire process is composed of simple rules, and is computationally not expensive.

3.2. Comparison with Literature

We can extend the approach in this project to a PROSA architecture framework[1] developed for product assembly. In these framework there are 4 types of holons:

- **Order:** Analyzes the requirements of a product's assembly and plans how to build it by using the resources available.
- **Product:** Keeps track of a product's individual requirements and evolution.
- **Resource:** Monitors a particular building resource (availability, state, etc).
- **Staff:** Guides the planning search process by giving suggestions on how to assemble a product.

In the sense of this project, we can make a parallel between: 1) the Resource holon and the Vehicle agent as the AGV is a resource of the system which is not always available, the Vehicle agent monitors this resource. 2) The Order holon and the CAA, as it plans how and when to use the resources (AGVs) in order to “build” the product, or in this context, transport the parcels. 3) The Product holon could derive the refinement of the parcels, as we could in principle assign an agent to monitor the parcel requirements and development. Currently it is arguably not an agent as it is not autonomous or independent, it requires triggering from agents. Additionally, it does not keep track of its own progress or requirements. Nevertheless, it could be easily turned into an agent. 4) Currently there is no equivalent for the Staff holon, nevertheless, it is mentioned that this is an “optional” structure, not indispensably required for the system to operate.

The full implementation of this parallel in this context could possibly yield a “recipe” for making the system more distributed and more efficient with a clear separation of concerns between the agents.

4. EXPERIMENTS

The purpose to this experiments is to prove or refute that the time for completely deliver a set of parcels to their destination can be reduced by the implementation of the energy function bidding. Additionally, the experimentation is conducted to prove the effectiveness of the agent-to-agent coordination for collision avoidance.

4.1. Setup

I propose to realize several experiments to attempt to tune the parameters of the energy function. Afterwards I obtain the results for the tuned energy equation which are compared with the results of a *normal allocation approach*: each AGV picks the closest parcel.

The former result is easy to obtain as it corresponds to the singular case where in Equation 1, β_1 , α_2 , and $\beta_2 = 0$, while $\alpha_1 = 1$.

Simultaneously, we examine the amount of times the simulation fails as a result of faulty agent-to-agent coordination for collision avoidance.

For the tuning of the energy function, I propose a rudimental approach where, 1) we systematically obtain results by setting all but one parameter to 0, while keeping that parameter equal to 1. 2) Afterwards, I weight the parameters based on their average performance compared to the total performance of all the combinations. This value is a rough approximation of the correct tuning. Nevertheless, this approximation might be sufficient to produce a better result than the *normal allocation approach*.

These experiments are realized by randomizing the initial position of the AGVs, and the parcels (pickup lo-

cation, delivery location, and expiry time) on every iteration. Nevertheless, this is a small limited grid, therefore the variation in distances do not fluctuate too much, we account for this by realizing multiple iterations and taking the average. In Figure 5, it is shown the experiment setup. Additionally, the number of AGVs (3) and the number of parcels (4) were chosen as it requires a good amount of coordination for the AGVs at the beginning to select which 3 parcels pick, and at the end, to select which AGV picks up the remaining parcel. Additionally the number is appropriate for the space of the environment.

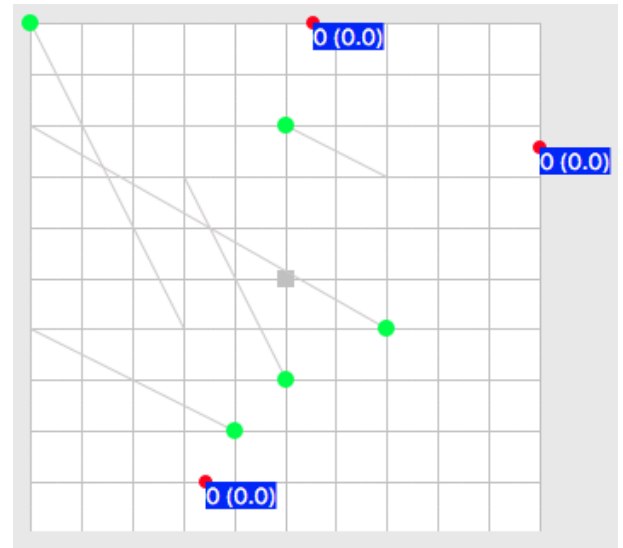


Figure 5: System setup. Red dots are AGVs, yellow dots are parcels, and the lines indicate the parcel’s delivery location. The center grey rectangle represents the CAA.

4.2. Results

All parameters set to 0 except the one mentioned on the table, see Table 2, Table 3, Table 4, and Table 5. The last value corresponds to the average. Note that Table 2 ALSO corresponds to the mentioned *normal allocation approach*.

$\alpha_1 = 1$
341000
216000
223000
195000
178000
262000
287000
348000
244000
212000
230600

Table 2: $\alpha_1 = 1$

$\beta_2 = 1$
341000
299000
213000
330000
279000
234000
248000
269000
296000
234000
292400

Table 5: $\beta_2 = 1$

The normalized and weighted parameters found where $\alpha_1 = 0.2601$, $\beta_1 = 0.2547$, $\alpha_2 = 0.2446$, and $\beta_2 = 0.2405$.

After imputing these parameters into the energy function and running the test again, I obtained the results in Table 6.

$\beta_1 = 1$
247000
324000
262000
199000
206000
382000
269000
270000
269000
227000
247600

Table 3: $\beta_1 = 1$

Tuned En- ergy F.
167000
262000
178000
245000
192000
292000
254000
300000
209000
227000
208800

Table 6: Results for the tuned energy function.

This accounts for an increase of 9.453% compared to the *normal allocation approach*. Additionally, it accounts for an increase of 15.6704% which respect to only taking into account the expiry time.

These results were obtain by leaving out the cases where the program failed due to a collision of AGVs or a deadlock. By running the program 20 extra times, we obtained that the program failed 8 times, for which the agent-to-agent coordination for collision avoidance failed 40% of the times.

4.3. Analysis

4.3.1 Energy Function Bidding

The experiment proved that even with a very rudimentary method for tuning the parameters in the energy

$\alpha_2 = 1$
268000
344000
234000
199000
352000
244000
262000
283000
240000
296000
279400

Table 4: $\alpha_2 = 1$

function, the results show a seizable improvement of $\approx 10\%$ over the *normal allocation approach*. Therefore by including in the allocation process, the distance and time information of all the parcels in range, the system is able to do a better allocation selection which corresponds to a decrease in the time for completion.

Nevertheless, this system can be improved in several ways. First, notice that the energy function increases value as the time increases. This is due to the time expiry parametrization, as in a long run, the time increases constantly, and so does energy functions. This could be improved by not taking into account only the expiry time but also the initial time, in other words, the size of the time window. This guarantees that the energy functions do not lose profitability over time.

Second, the method for tuning the parameters of the energy function produces a very rough approximation. For this approach I assume that each parameter is independent of the others, which is most likely not the case. Additionally I assume that the optimal parameters are normalized, which is also probably incorrect. More advanced methods for tuning could result in even better results. As a proposed method is Evolutionary Algorithms, which is a process that can be used for tuning unknown parameters. On the same branch, also Simulated Annealing could produce better results.

Third, on the same line of thought proposed on Section 3.2, we could use an Ant-based coordination and control by treating the parcels as products, the AGVs as resources and the CAA as order agents. The implementation of the energy function as proposed in this project is particularly useful for ants exploration as it relates picking a parcel now with the possibility of picking other parcels after delivery. Note, however, that this approach is useful where there are more parcels than AGVs, as in the converse case, the parcels planned to be available after delivery will likely be gone after delivery (see the footnote¹).

4.3.2 Agent-to-Agent Coordination for Collision Avoidance

On the other hand, the agent-to-agent coordination for collision avoidance performed poorly. **This can be due to the lack of a *control subsystem* that coordinates which layer, of the Practical Reasoning or Reactive layer, has control of the system. This**

results in cases where both systems are attempting to perform actions, which leads to unexpected behavior, and ultimately a collision. This does however does not falsify my hypothesis that this approach is effective; it can possibly be effective by implementing the mentioned control subsystem — more evidence is required. An implementation of such a system is a proposed improvement to this system.

Additionally, another possible improvement could be using an agent to perform the role of central authority to validate and control the conflicting plans, nevertheless, overdoing this might result into deprivation of the autonomy of the agents.

5. CONCLUSIONS

In this project I proposed and implemented a solution for dynamic task allocation based on a proposed energy function. Through the use of a central agent, I was able to guide the agents to the task allocation which profited the system the most — in terms of the energy function. This energy function is highly parametrized and can be controlled easily by modifying the values of the parameters. This can result in different behavior from the agents. By properly tuning this energy function to optimize the time the agents took to deliver all parcels, I was able to improve on the results obtained using a traditional allocation approach (the closest agent picks the parcel). The increase in efficiency was $\approx 10\%$. Additionally, I also designed and implemented a Planning Prior to Coordination system with agent-to-agent coordination for AGV collision avoidance, however the results obtained ranked lower than expected.

REFERENCES

- [1] H. V. Brussel et al. Reference architecture for holonic manufacturing systems: Prosa. *Computers in Industry*, 1998.
- [2] E. Durfee and S. Zilberstein. *Multiagent Planning, Control, and Execution*, chapter Chapter 11, Section 3. MIT Press, 2013.
- [3] J. Philips. *Holonic Task Execution Control of Multi-Mobile-Robot Systems*. PhD thesis, KU Leuven, 2012.

¹For these cases we can introduce some learning to the system, where the agents learn in which areas usually parcels appear more often, the energy function can then be altered to favor deliveries of parcels in areas where new parcels are more likely to appear. However, this is a different project.